

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE AND SYNTHESIS TRENDS

1. ABSTRACT

The supplied patent set spans a broad rubber-processing landscape, but only a subset directly addresses nitrile-containing elastomers and low-acrylonitrile NBR-relevant chemistry. Across the evidence, the target material is best understood as nitrile rubber systems with reduced acrylonitrile content or with formulations/processes that preserve oil resistance while improving processability, filler compatibility, low-temperature behavior, or downstream crosslinking performance. The most directly relevant disclosures include classical emulsion and graft-rubber routes (US2802808A), nitrile-rubber composition and cure packages for low rolling resistance and silica reinforcement (US6562929B2, US8785530B2), hydrogenated nitrile rubber with reduced molecular weight for improved processability (US7951875B2), and modern HNBR-based dispersant applications that explicitly specify acrylonitrile content windows (US11038175B2). Additional patents show adjacent but less direct approaches: controlled emulsion polymerization with in-situ emulsifier formation and block architectures (US7592409B2), one-pot liquid polymer/nanoparticle synthesis for rubber compounding (US7829624B2), and emulsion/anionic polymerization methods for structured polymers (EP3191546B1). Several patents in the manifest are outside the core NBR chemistry but illustrate broader formulation trends, including high-solids latex processing (US20240026191A1), sulfur-modified chloroprene cure optimization (EP3715409B1), and dispersant design for conductive materials (EP3950108B1).

The dominant polymerization approaches in the relevant NBR/HNBR patents are emulsion polymerization for nitrile-containing elastomers, solution polymerization for controlled molecular-weight copolymers, and post-polymer modification/hydrogenation to tune viscosity and performance. Monomer-content trends emphasize balancing acrylonitrile with butadiene and, in some cases, styrene or dicarboxylic-acid-derived units to manage oil resistance, compression set, and filler interaction. Formulation trends include use of polyamine crosslinkers, anti-aging agents, silica/carbon black, surfactant systems, seed latexes, and staged monomer addition. Overall, the evidence supports low-acrylonitrile NBR design as a tradeoff space: lower AN improves flexibility and processability, while cure package selection, hydrogenation, and staged polymerization are used to recover mechanical strength and chemical resistance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US6812307B2
- Patent Title: Functionalized monomers for synthesis of rubbery polymers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2004
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 3

- Patent Number: US7829624B2
- Patent Title: One-pot synthesis of nanoparticles and liquid polymer for rubber applications
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 4

- Patent Number: US10336853B2
- Patent Title: Polymer, process and composition
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 5

- Patent Number: US20240026191A1
- Patent Title: Styrene butadiene rubber latex compositions and methods for making and using ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 6

- Patent Number: US6562929B2

- Patent Title: Conjugated diene-based rubber and method of producing the same, oil extended ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2003
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: US8785530B2
- Patent Title: Crosslinkable rubber composition and cross-linked rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 8

- Patent Number: EP3715409B1
- Patent Title: Sulfur-modified chloroprene rubber composition, vulcanized product, molded ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 9

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 10

- Patent Number: US11038175B2
- Patent Title: Positive electrode active material pre-dispersion composition including ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 11

- Patent Number: US10597524B2
- Patent Title: Methacrylic resin, methacrylic resin composition, film, and production method
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 12

- Patent Number: US2802808A
- Patent Title: Compositions of a styrene-acrylonitrile resin, a graft of styrene and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1957
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 13

- Patent Number: EP3950108B1
- Patent Title: Dispersant, dispersed material, resin composition, mixture slurry, electrode ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 14

- Patent Number: US9670304B2
- Patent Title: Composite for preventing ice adhesion
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 15

- Patent Number: EP3191546B1
- Patent Title: Emulsion polymers including one or more 1,1-disubstituted alkene compounds, ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US6562929B2 vs US8785530B2: both center on nitrile-rubber-based vulcanizates for automotive/seal applications, but US6562929B2 specifies a conjugated diene-based rubber with 9-30 wt% nitrile, 10-50 wt% aromatic vinyl, and 20-81 wt% diene, whereas US8785530B2

focuses on highly saturated nitrile rubbers containing α,β -ethylenically unsaturated dicarboxylic acid monoester units and substantially no methacrylonitrile in one embodiment.

- US6562929B2 vs US7951875B2: US6562929B2 is a conventional nitrile-containing diene copolymer with staged nitrile addition during polymerization, while US7951875B2 uses hydrogenation and metathesis-derived molecular-weight reduction to obtain lower-MW HNBR with MW 30,000-250,000 and Mooney 3-50.
- US7951875B2 vs US11038175B2: both involve HNBR, but US7951875B2 targets low-MW HNBR feedstock for shaped articles, whereas US11038175B2 uses HNBR as a dispersant and specifies acrylonitrile-derived units at 20-50 wt% and hydrogenated butadiene ratio of 10-25% (with a broader discussion of 1-30% in the description).
- US7592409B2 vs EP3191546B1: both disclose controlled emulsion/anionic polymerization concepts, but US7592409B2 emphasizes in-situ emulsifier formation and controlled free-radical block copolymers including acrylonitrile/methacrylonitrile with isoprene, while EP3191546B1 uses micellar anionic emulsion polymerization of 1,1-disubstituted alkenes followed by sequential monomer addition to form block copolymers.
- US7829624B2 vs US20240026191A1: both are staged, multi-step rubber processes, but US7829624B2 forms liquid polymer plus in-situ nanoparticles in one batch using quenching and subsequent aromatic crosslinking, whereas US20240026191A1 uses seed-assisted, two-stage SBR latex growth to raise solids content and control zeta potential.
- US8785530B2 vs EP3715409B1: both are cure/formulation patents rather than primary polymerization patents; US8785530B2 uses polyamine crosslinking for nitrile/acrylic rubbers, while EP3715409B1 uses sulfenamide and xanthogen disulfide modification in sulfur-modified chloroprene latex.
- US2802808A vs modern nitrile patents: US2802808A is an early graft-copolymer latex route using butadiene rubber, styrene, and acrylonitrile with emulsifiers, persulfate-type initiators, and mercaptan chain transfer, foreshadowing later staged latex and graft strategies.
- US10597524B2 and EP3950108B1 are adjacent rather than core NBR patents: both use nitrile- or vinyl-cyanide-containing polymers in high-performance resin/dispersant contexts, showing that nitrile functionality is also leveraged outside elastomer synthesis for thermal stability and dispersion control.
- Across the set, recurring themes are staged monomer addition, surfactant/emulsifier control, and property balancing through composition rather than a single monomer ratio; newer patents more often specify narrow molecular-weight or solids windows, while older patents emphasize broad compositional ranges and graft/latex architecture.

4. CONCLUSION

The evidence indicates that low-acrylonitrile NBR development is less about a single monomer recipe and more about balancing acrylonitrile reduction with process and cure engineering. The most relevant synthesis guidance is to use emulsion or solution routes with staged monomer addition, controlled initiator/surfactant systems, and, where needed, hydrogenation or post-modification to recover oil resistance, compression-set performance, and processability. For

low-AN NBR-like targets, the strongest supported compositional window in the supplied data is a nitrile-containing diene copolymer with roughly 9-30 wt% nitrile content, optionally paired with aromatic vinyl comonomer and/or dicarboxylic-acid monoester functionality, followed by cure packages based on polyamines or tailored sulfur-modifier systems depending on the end use. If lower viscosity and improved handling are priorities, the HNBR route with reduced molecular weight is supported; if dispersion or battery-slurry use is the target, HNBR with about 20-50 wt% acrylonitrile-derived units and moderate hydrogenation is preferred. Overall, the patent evidence supports staged polymerization, careful emulsifier/initiator selection, and downstream crosslinking or hydrogenation as the most robust parameters for synthesizing and applying low-acrylonitrile NBR materials.

5. REFERENCES

- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- US6812307B2 | Functionalized monomers for synthesis of rubbery polymers | Not disclosed | US | 2004 | <https://patents.google.com/patent/US6812307B2/en>
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- US20240026191A1 | Styrene butadiene rubber latex compositions and methods for making and using ... | Not disclosed | US | 2024 | <https://patents.google.com/patent/US20240026191A1/en>
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- US8785530B2 | Crosslinkable rubber composition and cross-linked rubber | Not disclosed | US | 2014 | <https://patents.google.com/patent/US8785530B2/en>
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- US11038175B2 | Positive electrode active material pre-dispersion composition including ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11038175B2/en>
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- US2802808A | Compositions of a styrene-acrylonitrile resin, a graft of styrene and ... | Not disclosed | US | 1957 | <https://patents.google.com/patent/US2802808A/en>
- EP3950108B1 | Dispersant, dispersed material, resin composition, mixture slurry, electrode ... | Not disclosed | EP | 2024 | <https://patents.google.com/patent/EP3950108B1/en>

- US9670304B2 | Composite for preventing ice adhesion | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9670304B2/en>
- EP3191546B1 | Emulsion polymers including one or more 1,1-disubstituted alkene compounds, ... | Not disclosed | EP | 2020 | <https://patents.google.com/patent/EP3191546B1/en>